

Voltage Graphs on $\mathbb{Z}_3$ and Their Derived Covers

In each pair below, the left figure is the base graph Γ with edges/loops labeled by elements of $\mathbb{Z}_3$; the right figure is the derived (covering) graph Γ^λ , with three fibers of three vertices each. Edge colors in the derived graphs indicate voltage:

— voltage 0 — voltage 1 — voltage 2

Case 1 of 4: Only loops labeled 1, 2

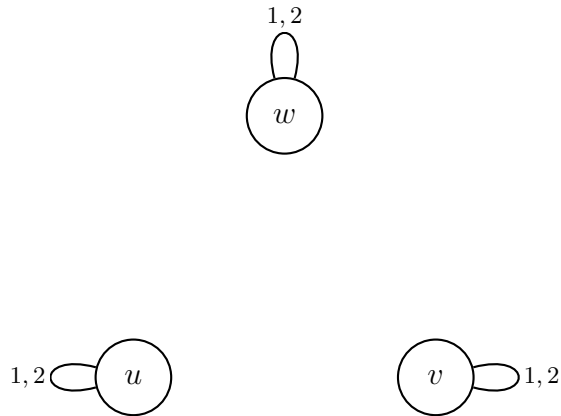


Figure 1: Base Γ : undirected loops

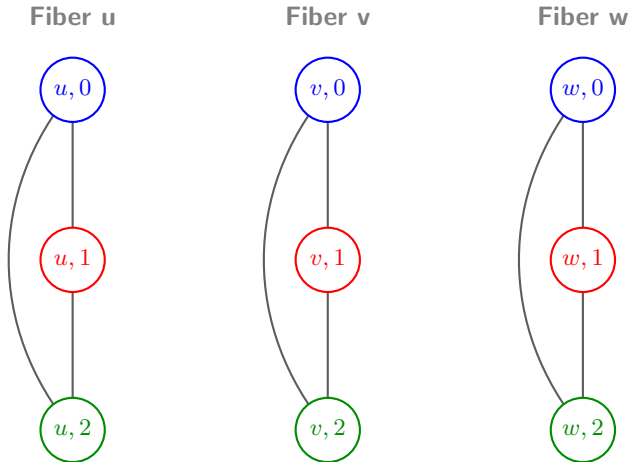


Figure 2: Derived Γ^λ : undirected 3-cycles

Case 2 of 4: Loops plus a 1-voltage triangle

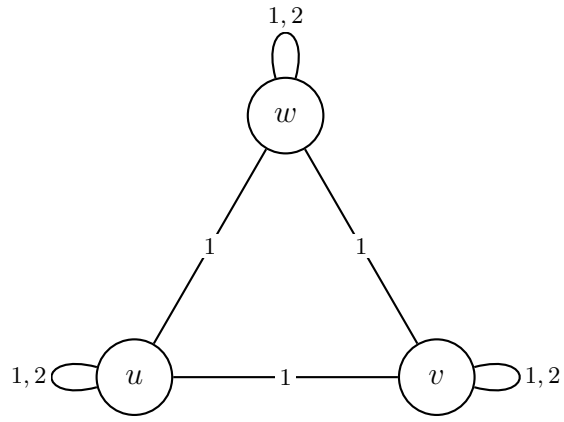


Figure 3: Base Γ : undirected triangle & loops

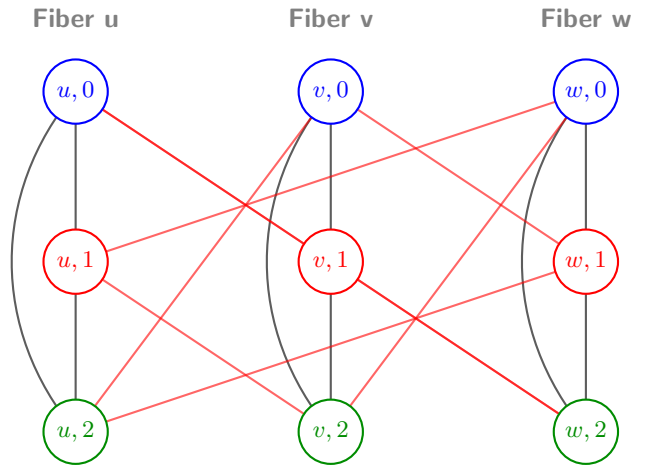


Figure 4: Derived Γ^λ (exactly 18 undirected edges)

Case 3 of 4: Loops plus 1- and 2-voltage triangle edges

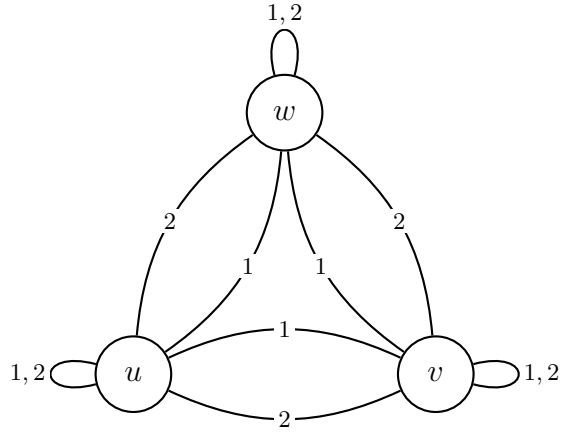


Figure 5: Base Γ : undirected multigraph

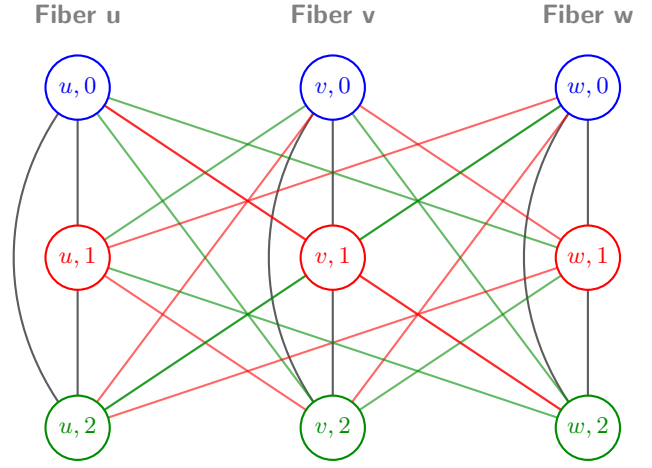


Figure 6: Derived Γ^λ (partial $K_{3,3}$ between fibers)

Case 4 of 4: Loops plus 0-, 1-, and 2-voltage triangle edges

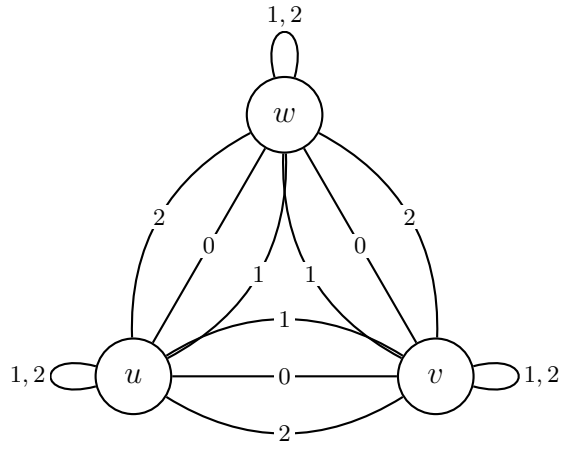


Figure 7: Base Γ : full undirected multigraph

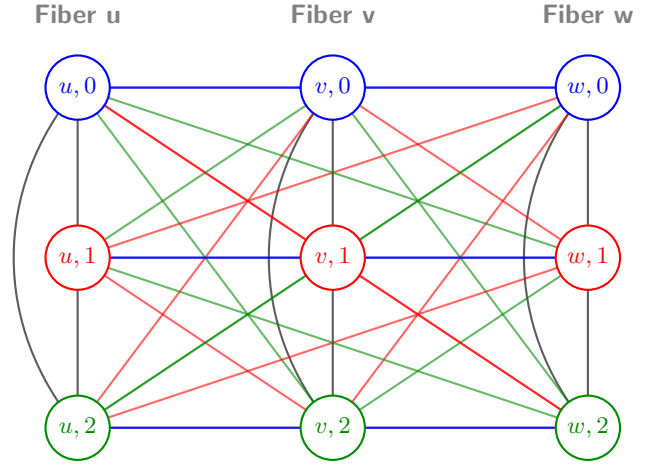


Figure 8: Derived Γ^λ (perfect $K_{3,3}$ between all fibers)